Supplementary Information
On the Conservation Laws and Pathway Realizability of Biochemical Reaction Networks

Thanh-An Pham[1] and Tieu-Long Phan[1,2,*]

[1]Bioinformatics Group, Department of Computer Science & Interdisciplinary Center for Bioinformatics & School for Embedded and Composite Artificial Intelligence (SECAI), Leipzig University, Härtelstraße 16–18, D-04107 Leipzig, Germany
[2]Department of Mathematics and Computer Science, University of Southern Denmark, DK-5230 Odense M, Denmark

*Corresponding author: tieu@bioinf.uni-leipzig.de
Thanh-An Pham: thanh-an.pham@uni-leipzig.de

S1 Python interface example

Code Example S1 shows network construction and analysis through `SynKit`'s Python interface. We use a 14-species glycolysis model with currency metabolites omitted to calculate CRNT invariants, conserved pools and minimal siphons, then verify that one glucose token reaches two pyruvates in 15 firings. Run the example from the source tree or unpacked source distribution. The research modules under `Experiment/CRN` are distributed separately from the installed runtime wheel.

```
from synkit.CRN import (build_native, native_backend, crnt_summary,
    conserved_moieties, find_siphons, siphon_persistence_details,
    PathwayRealizability, crn_to_sbml)
from Experiment.CRN.studies import build_kegg_crn, GLYCOLYSIS_FLUX

library = build_native("/tmp/synkit-crn", with_semiflows=True)
with native_backend(library):
    crn = build_kegg_crn("M00001", drop_currency=True)
    crn.n_species, crn.n_reactions
    report = crnt_summary(crn)
    report.deficiency, report.is_weakly_reversible
    len(conserved_moieties(crn)), len(find_siphons(crn))
    siphon_persistence_details(crn).persistence_ok

    check = PathwayRealizability().load_syncrn_and_flow(
        crn, flow=GLYCOLYSIS_FLUX,
        initial_marking={"alpha-D-Glucose": 1},
        species="label", reaction="label")
    check.build_petri_net_from_flow()
    outcome = check.realizability_result()
    outcome.status.value, len(outcome.certificate)
    sbml_document = crn_to_sbml(crn)
```

Code Example S1 Python analysis of reduced `KEGG` glycolysis (`M00001`): network construction, exact structural analysis, discrete pathway verification, and `SBML` export.

S2 Reference-network validation and numerical comparison

Table S1 lists the sources, validation objectives, and conservation-law residuals for the twelve reference networks.

Table S1 Sources, validation objectives, and conservation-law residuals for the reference reaction networks. SVD residual reports $\max|\mathbf{S}^T B|$ for a floating-point basis B. Rounded invalid counts columns of round$(1000B)$ with non-zero integer residuals. Exact residual evaluates `SynKit`'s integer basis. The open reactor has a zero-dimensional conservation space.

Network model	Source	Targeted property	SVD res.	Rounded inv.	Exact res.
Reversible isomerization	Feinberg [1], Ch. 3	Minimal weakly reversible, Deficiency Zero Theorem (positive branch)	1.1e-16	0	0
Irreversible isomerization	Feinberg [1], Ch. 3	Deficiency Zero Theorem (negative branch)	0.0e+00	0	0
Isomerization cycle	Feinberg [1], Ch. 3	Weak reversibility without individual reversible steps	2.2e-16	0	0
Three disjoint reversible pairs	Constructed test network	Disconnected linkage classes, with minimal semiflow supports isolating disjoint pairs rather than their union	1.1e-16	0	0
Disjoint pairs, non-unit stoichiometry	Constructed test network	Non-unit stoichiometry $E \rightleftharpoons 2F$: primitive conserved pool $2E + F$	1.1e-16	0	0
Michaelis-Menten	Michaelis & Menten, Feinberg [1]	Enzyme mechanism with two conserved pools, $\delta = 0$, non-weakly reversible	2.2e-16	1	0
Reversible Michaelis-Menten	Feinberg [1]	$\delta = 0$, with weak reversibility restored	1.7e-16	1	0
Edelstein bistable switch	Edelstein [2]	Minimal reversible $\delta = 1$ network with multiple steady states	0.0e+00	0	0
Futile cycle (single-site)	Wang & Sontag [3]	Distributive cycle with three independent pools	3.3e-16	3	0
Futile cycle (two-site)	Wang & Sontag [3]	Exact conservation in a multistationarity motif	5.3e-16	0	0
Shinar-Feinberg ACR motif	Shinar & Feinberg [4]	ACR motif, where a minimal siphon contains no P-semiflow support	1.1e-16	0	0
Open reactor ($0 \to A \to B \to 0$)	Constructed test network	Zero-complex handling, where cycle closure changes weak reversibility	0.0e+00	0	0

S2.1 Algebraic consistency and deficiency validation

We cross-checked deficiency using two formulations. With species-by-complex matrix $\mathbf{Y}$ and directed complex-graph incidence $\mathbf{I}_a$, we have $\mathbf{S} = \mathbf{Y}\mathbf{I}_a$ and

$$\delta \;=\; n - \ell - \operatorname{rank}(\mathbf{S}) \;=\; \operatorname{rank}(\mathbf{I}_a) - \operatorname{rank}(\mathbf{Y}\mathbf{I}_a). \tag{1}$$

Both formulations give the same deficiency for every benchmark, checking complex partitioning, exact rank computation, and stoichiometric assembly. Fig. S1 shows the cross-check and deficiency-theorem results for the Edelstein network.

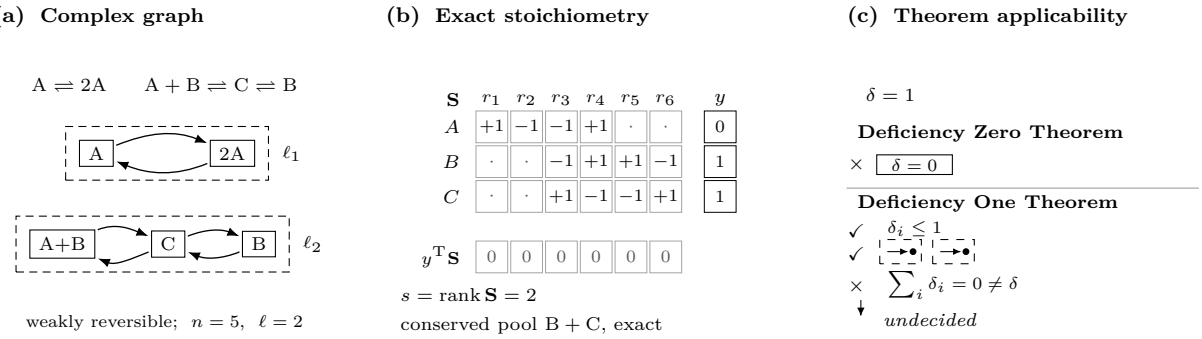


Figure S1 Conservation structure and deficiency-theorem checks for the Edelstein network [2]. **(a)** Complex graph and its two linkage classes. **(b)** Stoichiometric matrix beside the primitive conservation vector y, whose product $y^{\mathsf{T}}\mathbf{S}$ vanishes over $\mathbb{Q}$ and certifies the conserved pool [B] + [C]. **(c)** Each classical deficiency theorem fails one hypothesis, leaving multistationarity undecided by deficiency alone.

S2.2 Numerical nullspaces and exact integer conservation laws

We compare numerical and exact conservation identities on the twelve reference networks (Table S1). We compute the floating-point basis B by applying `scipy.linalg.null_space` to $\mathbf{S}^T$, with residuals ranging from zero to 5.3×10^{-16}. Small floating-point residuals do not ensure that scaling and rounding preserve an exact integer identity. The heuristic round$(10^3 B)$ produces non-zero exact residuals in Michaelis–Menten, reversible Michaelis–Menten, and the one-site futile cycle. This finding concerns that specific scaling and rounding choice. `SynKit`'s rational elimination returns the expected basis dimension, including zero for the open reactor, and every integer vector satisfies $y^T \mathbf{S} = 0$ exactly.

S3 Computational cost across network families

We measure five tasks at four sizes in three network families (Fig. S2), giving 60 completed measurements. Fig. S3 shows 48 measurements for structural analyses. The twelve standalone exact-rank measurements are omitted from this figure. Reversible-chain rank timings appear in Fig. 3c.

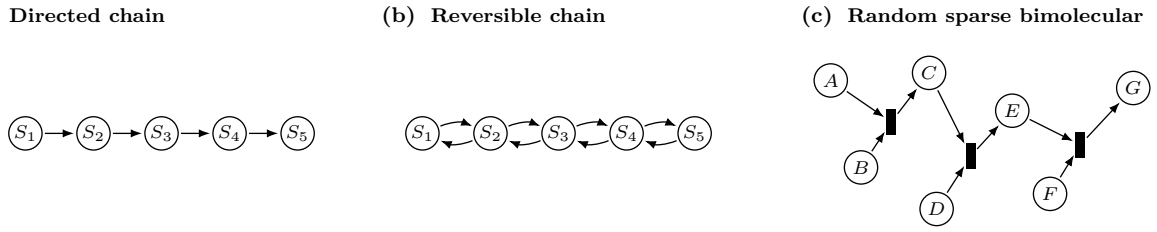


Figure S2 Network families used in the scaling benchmark. **(a)** Directed chain with one-way conversion steps. **(b)** Reversible chain with paired forward and reverse steps. **(c)** Random sparse bimolecular network with two substrates per reaction and sparse connectivity. Circles denote species and bars denote reactions.

S4 BioModels: structural SBML interoperability

The size-stratified selection in Table S2 contains ten reaction-bearing `BioModels` entries and the reaction-free control `BIOMD0000000562`. All ten reaction networks pass export and re-import checks without loss of integer stoichiometric multiplicities. The independent `sbml_reference.py` in `Experiment/CRN/studies` computes eight shared CRNT quantities from XML using only the Python standard library and exact rational arithmetic. We compare its results with the public importer under the retained, canceled and modifier-added readings of the reactions. We verify inputs against the packaged SHA-256 manifest. Reversible reactions expand into directed steps. Omitted reversibility flags default to true in Level 2. In Level 3 they trigger a strict error or lenient warning. Enumeration has a 1 000 000-node budget. Incomplete searches are flagged and excluded from completed counts and timings.

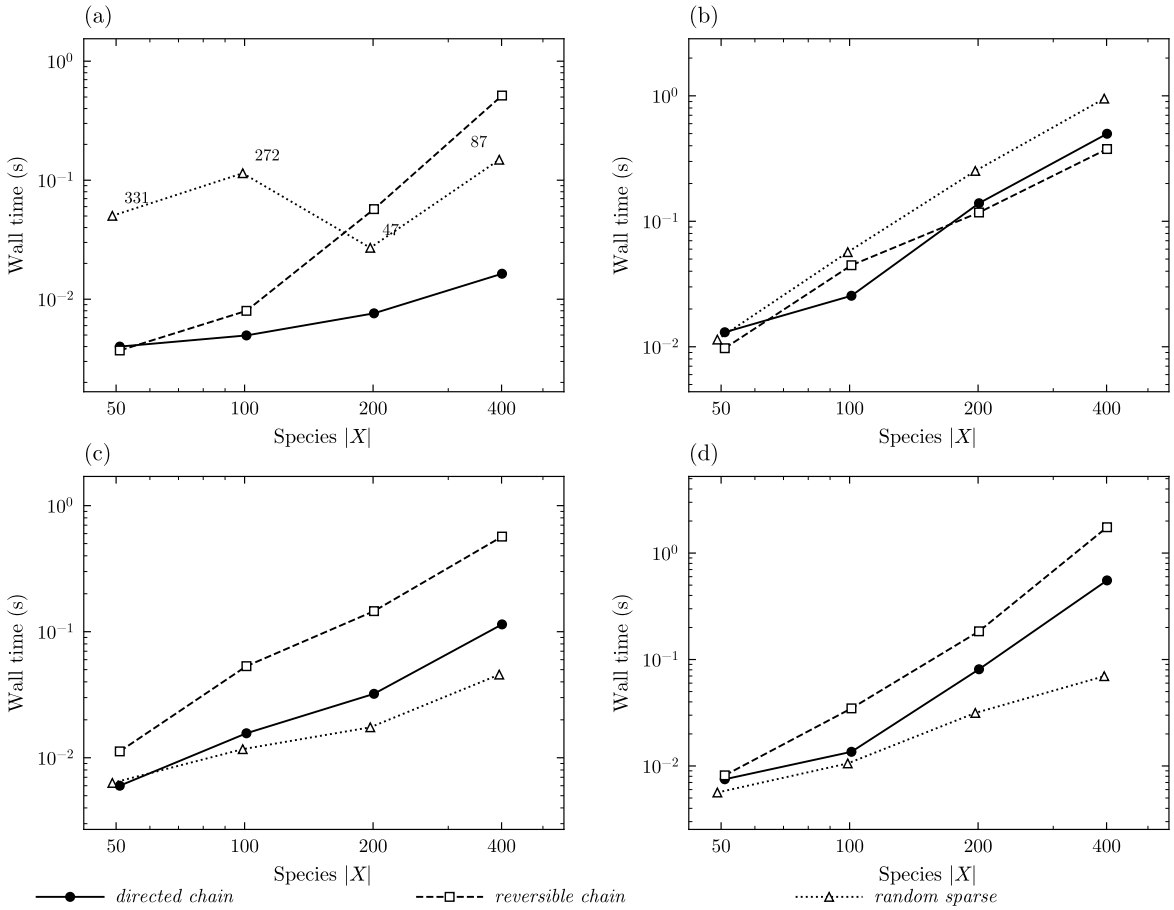


Figure S3 Single-run structural analysis runtimes for the three network families. **(a)** Minimal-siphon enumeration. Annotations give random-sparse siphon counts. **(b)** Exact CRNT analysis. **(c)** Graph canonicalization. **(d)** Exact conserved-pool enumeration. Line styles and markers distinguish families.

Table S2 BioModels results computed with SynKit version 1.7.0. m: species, r: directed reactions, s: exact rank, δ: deficiency, P: minimal conserved pools, S: minimal siphons, RT: supported structural round trip, Covered: every minimal siphon contains a conserved-pool support (vacuously true with no siphons). Coverage failure does not prove extinction. Dashes mark inapplicable reaction-free analyses. Reports record completion status.

Model	m	r	s	δ	P	S	RT	Covered
BIOMD0000000001	12	34	11	0	1	1	yes	yes
BIOMD0000000010	8	10	5	0	3	3	yes	yes
BIOMD0000000051	18	96	18	8	0	0	yes	yes
BIOMD0000000064	26	28	17	0	11	7	yes	no
BIOMD0000000108	9	18	7	8	2	2	yes	yes
BIOMD0000000404	29	60	19	13	10	10	yes	yes
BIOMD0000000637	12	35	9	9	3	3	yes	yes
BIOMD0000000019	100	242	81	41	19	19	yes	no
BIOMD0000000175	120	194	94	6	30	145	yes	no
BIOMD0000000255	504	1654	486	317	17	23	yes	no
BIOMD0000000562	0	0	–	–	–	–	–	–

S5 KEGG: module structure and glycolysis controls

Table S4 summarizes structural deficiency and conserved pools in glycolysis (M00001), pyruvate oxidation (M00307), citrate cycle (M00009), and pentose phosphate (M00004). The conserved pools recover nucleotide pairs, quinones, ferredoxins, and coenzyme A without manual curation.

Table S3 Catalyst conventions in the `BioModels` entries. Mod.: number of `SBML` modifier references, which the reader does not import. δ is the deficiency with species listed on both sides kept (as imported), with those species canceled as in net-stoichiometry tools, and with every modifier added to both sides of its reaction. The reaction-free control `BIOMD0000000562` is omitted. Input files are those of Table S2, checked by SHA-256.

Model	Mod.	δ kept	δ cancelled	δ with modifiers
`BIOMD0000000001`	0	0	0	0
`BIOMD0000000010`	5	0	0	5
`BIOMD0000000051`	5	8	8	13
`BIOMD0000000064`	10	0	0	0
`BIOMD0000000108`	2	8	8	8
`BIOMD0000000404`	2	13	11	15
`BIOMD0000000637`	0	9	9	9
`BIOMD0000000019`	0	41	41	41
`BIOMD0000000175`	142	6	6	8
`BIOMD0000000255`	0	317	317	317

Table S4 Structural results for four reduced KEGG modules. m: species, r: directed reactions, n: complexes, ℓ: linkage classes, s: rank, $\delta = n - \ell - s$: deficiency, P: minimal conserved pools, and S: minimal siphons.

Module	m	r	n	ℓ	s	δ	P	S
`M00001`	14	15	16	4	11	1	3	3
`M00307`	13	8	14	7	6	1	8	6
`M00009`	22	12	23	11	12	0	12	9
`M00004`	12	12	14	4	10	0	2	4

S5.1 Glucose-to-pyruvate pathway realization and invariants

Table S5 specifies the reaction multiset converting one glucose to two pyruvates. Preparatory reactions fire once. Aldolase (`R01068`) and triosephosphate isomerase (`R01015`) supply two glyceraldehyde 3-phosphate equivalents, so each payoff reaction fires twice. State-space search yields a certified 15-step sequence (Fig. 5b). Every firing preserves the weighted three-carbon backbone, $2 \times \text{hexoses} + 1 \times \text{trioses} = 2$.

Table S5 Non-zero firing counts of the prescribed KEGG glycolysis flux. Identifiers match `M00001` reaction records.

Reaction	Count	Reaction	Count
`R01786`	1	`R01061`	2
`R13199`	1	`R01512`	2
`R00756`	1	`R01518`	2
`R01068`	1	`R00658`	2
`R01015`	1	`R00200`	2

S5.2 Generic compound identifiers and cofactor boundary controls

In polyphosphate glucokinase (`R02189`, EC 2.7.1.63), `KEGG` uses the generic identifier `C00404` for both $(\text{Phosphate})_n$ and $(\text{Phosphate})_{n-1}$, writing `C00404 + C00267 <=> C00404 + C00668`. Chemically, one phosphate moves from the polymer to glucose, so the polymer is a phosphoryl donor. `SynKit` retains the equation literally, making the polymer token unchanged in the reduced model. This generic-identifier effect therefore differs from the catalyst retention illustrated by `BIOMD0000000404` (Table S3).

In Table S6, we keep the prescribed multiset fixed and start from one glucose token with all other species absent. Retaining currency metabolites leaves hexokinase without ATP and no requested reaction enabled.

Removing them permits the 15-step carbon-backbone certificate. The negative result applies to this multiset and initial inventory.

Table S6 Currency-boundary control for the prescribed full glycolytic flux from one glucose and zero tokens of all other species. The negative decision is exhaustive for this reaction multiset.

Currency metabolites	Species	Reactions	Decision	Certificate length
Removed	14	15	realizable	15
Retained	24	15	unrealizable	–

S5.3 Initial cofactor inventories and firing order

Table S7 holds the full prescribed reaction multiset fixed and varies only the initial inventory. From glucose alone (I_0), the adenylate set $\{\mathrm{ATP}, \mathrm{ADP}\}$ is an empty siphon in the selected reaction subnet, which rules out ATP-dependent startup. Supplying ATP alone (I_1, I_2) still leaves net deficits of two ADP, two NAD^+ and two orthophosphate tokens. Adding NAD^+ removes only its deficit (I_3). Adding phosphate as well still leaves an ADP deficit (I_4). These obstructions follow already from $m_0 + \mathbf{S}f \not\geq 0$.

Inventory I_5 supplies one ATP, two ADP, two NAD^+ and two orthophosphate tokens alongside glucose. Its predicted final inventory is non-negative, but the prescribed multiset is unrealizable because two ATP-consuming preparatory steps must occur before any ATP-producing payoff reaction. Exhaustive search stops after three markings. With two initial ATP instead (I_6), an independently replayed 15-step certificate completes the same multiset after exploring 30 markings. The final inventory is two pyruvates, four ATP, two NADH, two water and two protons, with no unlisted tokens. No case receives replenishment during firing.

Table S7 Initial-inventory controls for the prescribed full glycolytic flux. Each case starts with one glucose and the listed cofactor tokens, with zero tokens of all other species. P_i denotes orthophosphate. Net tests $m_0 + \mathbf{S}f \geq 0$, States counts explored markings, and Length is the verified certificate length. Negative decisions are exhaustive for the prescribed multiset.

Case	ATP	ADP	NAD^+	P_i	Net	Decision	States	Length
I_0	0	0	0	0	no	unrealizable	1	–
I_1	1	0	0	0	no	unrealizable	3	–
I_2	2	0	0	0	no	unrealizable	6	–
I_3	2	0	2	0	no	unrealizable	6	–
I_4	2	0	2	2	no	unrealizable	27	–
I_5	1	2	2	2	yes	unrealizable	3	–
I_6	2	2	2	2	yes	realizable	30	15

S6 Formose networks: expansion and precursor requirements

Formose expansion starts from formaldehyde (C_1) and glycolaldehyde (C_2), using aldol coupling and enediol-mediated keto-enol tautomerization.

S6.1 Network growth and stoichiometric carbon conservation

After four expansion iterations (Table S8), the network contains 37 species and 44 reactions and reaches species with ten carbon atoms (C_{10}). Exact nullspace analysis identifies a single primitive positive conser-

vation semiflow. Its weight for every species equals that species' carbon atom count, confirming carbon conservation across all generated reactions.

S6.2 Comparative rule expansion benchmarking

We compare full API runtimes for `SynReactor` DPO, `MØD`, and `MolFoundry` in Table S8. Each tool runs on one logical CPU from fresh formaldehyde and glycolaldehyde seeds using its native rule format. We canonicalize molecules without explicit hydrogens before comparing outputs. All three tools produce identical species and directed stoichiometric reaction sets at each of the six expansion depths.

Table S8 Full API runtime comparison. Times are median cumulative seconds from seeds over 20 timed runs after three warm-ups, rounded to three significant figures as in the main text. Native `SynReactor` uses its C++ controller and DPO kernel with implicit hydrogen counts, whereas `MØD` and `MolFoundry` use explicit hydrogen vertices. Rule preparation and network construction are included for every tool. Counts describe identical completed species and stoichiometric reaction sets. Each expansion has a 180 s time limit and an 8 GiB per-process address-space limit.

Depth	Species	Reactions	`SynReactor` C++ DPO	`MØD`	`MolFoundry`
1	3	1	0.0275	0.000705	0.0031
2	5	4	0.0326	0.000947	0.00634
3	9	10	0.0272	0.00152	0.00793
4	37	44	0.0283	0.00369	0.0229
5	302	371	0.0642	0.0362	0.26
6	10572	11239	1.86	2.22	15.1

S6.3 Route definitions and finite precursor inventories

The targets are an aldotetrose (T_4, `O=CC(O)C(O)CO`) and an aldohexose (T_6, `O=CC(O)C(O)C(O)C(O)CO`). Under qualitative set reachability, both are reachable from the two seed species. Table S9 distinguishes qualitative depth from the reaction multiplicities required for discrete firing, and Fig. S4 shows the reaction motifs.

Table S9 Qualitative depths and prescribed discrete reaction multisets for the selected formose targets. Set reachability reuses a species once generated, whereas discrete firing requires a separate token for each consumed molecule.

Target	Qualitative depth	Discrete firings	Prescribed multiset
T_4	2	2	$r_1 + r_3$
T_6	3	4	$2r_1 + r_3 + r_7$

We test both target routes from five initial precursor inventories (Table S10). For an independent comparison, we supply the same prescribed reaction multisets and initial markings to `MolFoundry`'s `FlowSolution`, without optimizing a new hyperflow. Its `is_realizable()` method and SMT certificate enumeration agree with all ten `SynKit` decisions. We independently replay all three positive certificates, checking enabling before every firing, the requested reaction multiplicities, and target production. Negative decisions come from exhaustive search over the prescribed multiset. They do not rule out other routes to the targets.

The tetrose route first converts glycolaldehyde to enediol (r_1), then couples it with a second glycolaldehyde (r_3). It therefore requires two glycolaldehydes and is supported by M_4 and M_5. The hexose route adds a second enediol through r_7, requiring three glycolaldehydes in total, which only M_5 supplies (Fig. S4). Neither route consumes formaldehyde. The equal-carbon inventories M_3 to M_5 differ in realizability be-

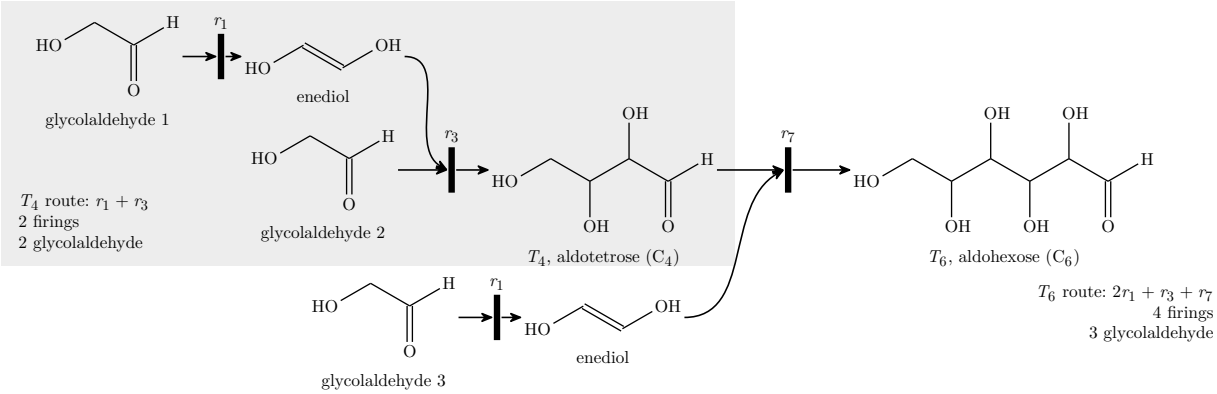


Figure S4 Precursor requirements of the selected formose routes. Bars are reactions, as in the Petri nets of the main text, and each glycolaldehyde is drawn as a separate molecule because discrete firing consumes one token per molecule. Keto–enol tautomerization (r_1) supplies the enediol consumed by each aldol addition (r_3, r_7). The shaded block is the T_4 route ($r_1 + r_3$: two firings, two glycolaldehyde), and the whole scheme is the T_6 route ($2r_1 + r_3 + r_7$: four firings, three glycolaldehyde). Structures are constitutional, with no stereochemistry or atom mapping assigned.

Table S10 Realizability of the prescribed tetrose and hexose routes across five precursor inventories. Counts denote molecules. Total carbon counts carbon tokens. Yes indicates a verified certificate and No an exhaustive negative decision for the selected multiset.

Pool	C_1 count	C_2 count	Total carbon	T_4	T_6
M_1	1	1	3	no	no
M_2	2	1	4	no	no
M_3	4	1	6	no	no
M_4	2	2	6	yes	no
M_5	0	3	6	yes	yes

cause they supply different numbers of the required precursor. We do not test alternative target-producing multisets from these inventories.

An independent structural obstruction applies to formaldehyde alone. An eight-species minimal siphon contains glycolaldehyde (`O=CCO`), its enediol (`OC=CO`), and both targets, but excludes formaldehyde (`C=O`). If all eight species are absent initially, no reaction can produce a member, so formaldehyde alone cannot initiate either target. All five tested inventories already mark this siphon through glycolaldehyde. Their selected-route decisions therefore require the separate multiplicity analysis above.

References

[1] Martin Feinberg. *Foundations of Chemical Reaction Network Theory*, volume 202 of *Applied Mathematical Sciences*. Springer, Cham, 2019. doi:10.1007/978-3-030-03858-8.

[2] Boris B Edelstein. Biochemical model with multiple steady states and hysteresis. *Journal of Theoretical Biology*, 29(1):57–62, 1970. doi:10.1016/0022-5193(70)90118-9.

[3] Liming Wang and Eduardo D Sontag. On the number of steady states in a multiple futile cycle. *Journal of Mathematical Biology*, 57(1):29–52, 2008. doi:10.1007/s00285-007-0145-z.

[4] Guy Shinar and Martin Feinberg. Structural sources of robustness in biochemical reaction networks. *Science*, 327(5971):1389–1391, 2010. doi:10.1126/science.1183372.